

SAMA MUSHTAQ

MECHANICAL DESIGN AND TEST TECHNICIAN

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Mechanical engineering training with 103 earned credits, one year of professional MEP design work across more than 50 building projects, and project experience in LabVIEW data acquisition, test setup, structural analysis, and heat transfer. Ten years of operations leadership added practical troubleshooting, clear documentation, and ownership from plan through result.

ENGINEERING AND TECHNICAL EXPERIENCE

- MKEC Engineering** Mechanical Design Engineer **Dec 2019 to Dec 2020**
- Prepared heating and cooling load calculations and HVAC, plumbing, electrical, equipment, and airflow schedules for more than 50 commercial, residential, and modular projects using Trane TRACE, AutoCAD, and Revit.
 - Built reusable Revit families, templates, and details that reduced design time by about 25 percent, and helped resolve field conflicts through site visits and multidisciplinary coordination.
- RYGNeco** Co Founder and Program Lead **Aug 2024 to present**
- Wrote the procedures and product requirements for electronics intake, testing, grading, chain of custody, disposition, evidence, finance, and client reporting.
 - Directed an approximately \$11,500 vendor product build through scope, milestones, acceptance criteria, and review. The field program produced a verified 130 device test record for a healthcare pilot.
- Mercor** AI Annotator and Quality Reviewer, Contract **Feb 2026 to present**
- Evaluate technical and business work against detailed rubrics, trace claims and calculations to supplied evidence, and review other contributors' work for accuracy and instruction following.
- MAK Trading** Founder and Research Systems Builder **Nov 2019 to present**
- Built a Python and SQLite research system with Codex that preserves original Discord trade calls and edits, records market context and outcomes, and keeps moderator corrections with the source record.

ENGINEERING PROJECTS

- **Vibration sensing and data acquisition 2016** Built and tested a small Keystudio and MiniSense sensor setup, captured 1,022 samples across three LabVIEW channels, and compared both responses in Excel.
- **Shell and tube heat exchanger 2016** Completed an individual countercurrent water to oil design study from heat duty through required area and tube length, comparing ten materials under clean and fouled conditions.
- **Kite buggy structural analysis 2015** Worked on a five person team that developed a portable 6061 aluminum frame weighing 29.7 pounds and evaluated it against the loads from a modeled ten foot drop case.

OPERATIONS LEADERSHIP

- **Hard Rock Casino Cincinnati 2024 to 2025** Led work across three food and beverage outlets with about \$17 million in combined operating scope and 120 employees, connecting inventory, labor, product mix, training, and manager review.
- **EPIC Brands and iTZCALi 2022 to 2025** Led staffing, training, private dining development, and new concept work at EPIC Brands after coordinating the opening of a 300 seat restaurant at iTZCALi.

EDUCATION AND TOOLS

University of Cincinnati Admitted to Mechanical Engineering for Spring 2027. Prior engineering credits are under transfer review.

Wichita State University Mechanical Engineering studies with 103 earned credits, a 3.4 GPA, and three Dean's List terms.

University of Wales Trinity Saint David Business and Management studies with 25 evaluated United States semester credits and a 3.47 GPA.

Design and test AutoCAD, Revit, Trane TRACE, LabVIEW, CATIA, Excel, test setup, data acquisition, technical documentation

Technical Python, SQL, GitHub, ChatGPT, Claude, Cursor, OpenAI Codex, n8n

Languages English, Urdu, and Hindi